NANYANG TECHNOLOGICAL UNIVERSITY

EE6222 Machine Vision

Assignment 1: Dimensionality Reduction for Classification

MSc in Electrical and Electronic Engineering
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October 20, 2025

Contents

1 Introduction

In this Assignment, I selected the MINIST dataset for training and testing the performance of classifiers. As for the dimension of images are largee, I explored several dimensionality reduction methods to reduce the dimension of data while trying to retain the most important features, including **PCA**, **LDA**, combined method (Fisherfaces), **LLE**, **Isomap**, and **PaCMAP**. I compared the effectiveness of these methods by plotting their performance with varying number of dimensions.

In addition, I establish several basic classifiers, like **MMD**, **KNN**, and include Logistic Regression Classifier with sklearn package. Therefore, I not only compare the effectiveness of different dimensionality reduction methods, but also access the accuracy of classifiy images for these classifer methods. I would describe the details of my experiment in the flowing sections.

2 Dataset and Preprocessing

2.1 Dtaset Descrption

The MINIST dataset contains **70,000** images of handwritten digits (0-9) with a resolution of 28x28 pixels, which means there are **10** classes intotal. Each image had been saved with the format of vector with **784 in length**. In addition, the images are only have one channel, which means they are grayscale pixels. Because of the pre-shaped vectors, single channel, and easy to get features, I think MINIST dataset is suitable for this assignment. Therefore, I fetch the dataset from sklearn package directly 1.

```
mnist = fetch_openml('mnist_784', version=1)
```

Figure 1: Fetch Code

2.2 Normalization and Standarlization

Since the MINIST dataset had been well preprocessed and tested by plenty of researchers, I didn't do too much basic preprocessing, like outlier and missing value detection. However, I though normalization and standarlization arer still neccessary for the following basic classifiers, which may be affected intensely by the scle of features. So I rearrange the vectors to 0 mean and unit variance.

Moreover, I found the number of samples (70,000) is too large, which may lead to many difficulties in my following experiment. Firstly, in the following experiment, I would traing several classifiers with many

3 Methodology

4 Results and Discussion

5 Conclusion

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References

[1] I. K. Fodor, "A survey of dimension reduction techniques," Tech. Rep. UCRL-ID-148494, Lawrence Livermore National Laboratory, Livermore, CA, USA, May 2002.